

Open Mathematical Olympiad for University Students (OMOUS-2022)

Individual Competition

Group A

Problem 1 Three types of bacteria A, B and C are placed into a closed container. Initially, there is 1 unit of A, 2 units of B, and 4 units of C bacteria. Every day, after growing to a certain size, each bacterium splits into 5 bacteria, 3 of which are of its own type, and 2 of which are of each of the other types. Find the amount of each type of bacteria after 2022 days, and the total number of bacteria.

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Let x_n, y_n and z_n are total number of bacteria's A, B and C respectively after n day.

Then we will have recurrence formula for x_n, y_n and z_n sequences:

$$\begin{cases} x_{n+1} = 3x_n + y_n + z_n \\ y_{n+1} = x_n + 3y_n + z_n \\ z_{n+1} = x_n + y_n + 3z_n \end{cases}$$

Let $S_n = x_n + y_n + z_n$, then $S_{n+1} = x_{n+1} + y_{n+1} + z_{n+1} = 5(x_n + y_n + z_n)$
So $S_n = 5^n S_0 = 7 * 5^n$

From here we get $x_{n+1} - 2x_n = 7 * 5^n$. Multiplying both sides to $2^{-(n+1)}$

$$\begin{aligned} \frac{x_{n+1}}{2^{(n+1)}} - \frac{x_n}{2^n} &= \frac{7}{2} \left(\frac{5}{2}\right)^n \\ \sum_{k=0}^{n-1} \left(\frac{x_{k+1}}{2^{(k+1)}} - \frac{x_k}{2^k}\right) &= \sum_{k=0}^{n-1} \frac{7}{2} \left(\frac{5}{2}\right)^k = \frac{7}{2} \frac{\left(\frac{5}{2}\right)^n - 1}{\frac{5}{2} - 1} = \frac{7 * (5^n - 1)}{3 * 2^n} \\ \frac{x_n}{2^n} - \frac{x_0}{2^0} &= \frac{7 * (5^n - 1)}{3 * 2^n} \end{aligned}$$

and

$$x_n = \frac{7}{3} (5^n - 1) + 2^n$$

Similarly we can calculate y_n and z_n

Problem 2 Let $\{a_n\}$ be a recursive sequence given by

$$a_0 = 1, a_{n+1} = \frac{2a_n}{2 + \sqrt{4 - a_n^2}}$$

Prove that $\{2^n a_n\}$ is increasing and find its limit.

Karen Keryan (Yerevan State University & American University of Armenia, Armenia)

Clearly, a_n is positive and decreasing, so $a_n \in [0, 1]$.

Let b_n be so that $a_n = 2 \tanh(b_n)$. Then we have

$$\begin{aligned} a_{n+1} &= \frac{4 \tanh(b_n)}{2 + \sqrt{4 - \tanh^2(b_n)}} = \frac{2 \sinh(b_n)}{\cosh(b_n) + \sqrt{\cosh^2(b_n) - \sinh^2(b_n)}} = \frac{2 \sinh(b_n)}{\cosh(b_n) + 1} \\ &= 2 \tanh\left(\frac{b_n}{2}\right). \end{aligned}$$

Repeatedly using this equality we get

$$a_n = 2 \tanh(b_n) = 2 \tanh\left(\frac{b_{n-1}}{2}\right) = \dots = 2 \tanh\left(\frac{b_1}{2^{n-1}}\right) = 2 \tanh\left(\frac{b_0}{2^n}\right).$$

Consider $f(x) = \frac{\tanh x}{x}$. Note that

$$f'(x) = \frac{\frac{x}{\cosh^2 x} - \tanh x}{x^2} = \frac{x - \sinh x \cosh x}{x \cosh^2 x} = \frac{2x - \sinh(2x)}{2x \cosh^2 x}.$$

Since $t - \sinh t$ is decreasing ($(t - \sinh t)' = 1 - \cosh t \leq 0$), it follows that $t - \sinh t \leq 0$ for $t \geq 0$. Hence $f'(x) \leq 0$ and f decreases.

Notice that $2^n a_n = 2b_0 f\left(\frac{b_0}{2^n}\right)$, so $2^n a_n$ increases. From $2 \tanh(b_0) = a_0 = 1$ we obtain that $b_0 = \frac{\ln 3}{2}$.

Hence

$$\lim_{n \rightarrow \infty} 2^n a_n = 2b_0 \lim_{x \rightarrow 0} f(x) = 2b_0 = \ln 3.$$

Problem 3 Let $A, B \in M_2(\mathbb{R})$ matrices such that

a) B is invertible

b) $A^2 + B^2 = 2AB$

c) $\det(A^d - B^d) = \det(A^d + B^d)$ for $d = \{2020; 2021\}$

Prove that $\det(A^{2022} - I_2) = \det(A^{2022} + I_2)$

Murat Chashemov (Turkmenistan)

Solution.

Lemma 1: If $A, B \in M_2(\mathbb{R})$ and $A^2 + B^2 = 2AB$, then $AB = BA$.

Proof: From condition of lemma

$$(A - B)^2 = A^2 + B^2 - AB - BA = AB - BA$$

Then

$$\text{Tr}(A - B)^2 = \text{Tr}(AB - BA) = 0$$

and

$$\begin{aligned} \text{Tr}(A - B)^3 &= \text{Tr}((A - B)(AB - BA)) = \text{Tr}(A^2B - ABA - BAB + B^2A) \\ &= \text{Tr}(A^2B) - \text{Tr}(ABA) - \text{Tr}(BAB) + \text{Tr}(B^2A) = 0 \end{aligned}$$

Let α_1 and α_2 are eigenvalues of $(A - B)$ matrix, then

$$\begin{cases} \alpha_1^2 + \alpha_2^2 = 0 \\ \alpha_1^3 + \alpha_2^3 = 0 \end{cases} \text{ or } \alpha_1 = \alpha_2 = 0$$

So we get $(A - B)$ matrix is nilpotent and $(A - B)^2 = O_2$. From here $AB = BA$.

By using lemma 1 and condition a) we have $AB^{-1} = B^{-1}A$. Multiplying both sides of condition c) to the B^{-d}

$$\det(A^d B^{-d} - I_2) = \det(A^d B^{-d} + I_2)$$

or

$$\det((AB^{-1})^d - I_2) = \det((AB^{-1})^d + I_2)$$

Let $C = AB^{-1}$ and has eigenvalues α and β , then we get

$$(\alpha^d - 1)(\beta^d - 1) = (\alpha^d + 1)(\beta^d + 1)$$

and

$$\alpha^d + \beta^d = 0 \text{ or } \alpha = \beta = 0$$

Taking consideration A and B^{-1} matrices are commute, we know matrices A and B has common eigenvector. Let x is common eigenvector of matrix A and B^{-1} and γ and μ are eigenvalues of A and B^{-1} .

$$AB^{-1}x = \gamma\delta x = 0 \Rightarrow \gamma = 0$$

So A matrix is nilpotent and $A^2 = O_2$

Then $\det(A^{2022} - I_2) = \det(A^{2022} + I_2) = \det(\mp I_2) = 1$

Problem 4 Let S_1, S_2 be two internally tangent circles on plane, S_1 interior to S_2 with radii R_1 and R_2 ($R_1 < R_2$) respectively. Let $Q_1, Q_2, Q_3, \dots, Q_{2021}, Q_{2022}$ be any sequence of circles tangent to S_1, S_2 and successively to the previous and next circle in the sequence (i.e. Q_k is externally tangent to S_1 , internally tangent to S_2 , and tangent Q_{k-1} and Q_{k+1} for all k). Finally let P_k be the point of tangency of the circles Q_{k-1} and Q_k . Prove that all points $P_1, P_2, P_3, \dots, P_{2022}$ lie on a circle and calculate its radius.

Marcin Zygmunt (University of Silesia, Poland)

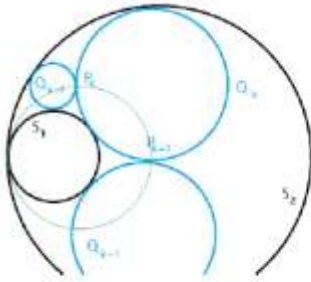


Fig. 1: Sequence of circles.

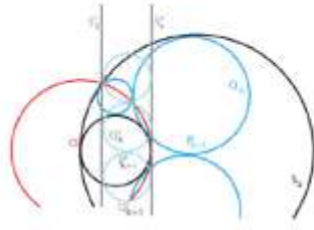


Fig. 2: Circles and their images through homography.

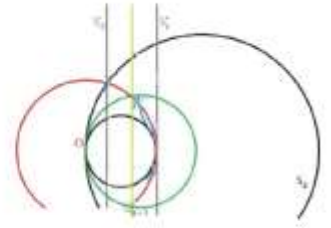


Fig. 3: Final circle

Solution. First, let us inscribe both circles S_1 and S_2 in the complex plane so that their point of tangency is at the origin of the coordinate system, i.e. at the point O . By a homography $z \rightarrow \frac{(2R_1)^2}{z}$ (i.e. a complex inversion) we can transform both circles S_1 and S_2 into a pair of parallel S_1' and S_2' lines distant from each other by $2R_1 \left(1 - \frac{R_1}{R_2}\right)$ (cf. fig.2). As all tangencies are preserved under inversion, the circles $Q_1', Q_2', Q_3', \dots, Q_{2021}', Q_{2022}'$ i.e. images of $Q_1, Q_2, Q_3, \dots, Q_{2021}, Q_{2022}$ are tangent to lines S_1', S_2' and corresponding circles from the sequence. That means the circles $Q_1', Q_2', Q_3', \dots$ have the same radius (cf. fig.).

The points $P_1', P_2', P_3', \dots$ lie on the line, parallel to S_1' and S_2' and intersecting the OX axis at a point distant from O by

$$d = \frac{1}{2} \left(\frac{1}{2} R_1 + \frac{(2R_1)^2}{2R_2} \right) = \frac{R_1}{R_2} (R_1 + R_2)$$

But complex homography transforms circles and lines to circles and lines, and preserves angles between them. Hence all points $P_1, P_2, P_3, \dots$ lie on the same circumference (cf. fig.), whose diameter is equal to

$$\frac{(2R_1)^2}{d} = \frac{4R_1 R_2}{R_1 + R_2}$$

i.e. whose radius equals to

$$\frac{2R_1 R_2}{R_1 + R_2}$$

Remark: The sequence of circles Q_n is called Steiner chain.

Problem 5 For positive integer n , let $S_1, S_2, \dots, S_{2^n-1}$ be the nonempty subsets of $\{1, 2, \dots, n\}$ in some order, and let M be the $(2^n - 1) \times (2^n - 1)$ matrix whose (i, j) entry is $m_{ij} = |S_i \cup S_j|$. Find the determinant of $M = (m_{ij})$.

Murat Chashemov (Turkmenistan)

Solution.

If $n = 1$, then $M = [1]$ and $\det(M) = 1$. Suppose $n \geq 2$. Interchanging S_i and S_j corresponds to switching the i th and j th columns and rows, thus leaves the determinant invariant. Hence, for $n = 2$, with $S_1 = \{1\}$, $S_2 = \{2\}$, and $S_3 = \{1, 2\}$, we get

$$\det(M) = \det \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 2 \end{bmatrix} = 2.$$

Now consider $n \geq 3$. Let $S_j = \{1\}$, $S_k = \{2, \dots, n\}$ (the complement of S_j), $S_p = \{2\}$, and $S_q = \{1, 3, \dots, n\}$ (the complement of S_p). Then for any $i = 1, \dots, 2^n - 1$, $|S_i \cap S_j| + |S_i \cap S_k| = |S_i|$, thus by PIE,

$$\begin{aligned} m_{ij} + m_{ik} &= |S_i \cup S_j| + |S_i \cup S_k| \\ &= |S_i| + 1 - |S_i \cap S_j| + |S_i| + (n - 1) - |S_i \cap S_k| = n + |S_n|. \end{aligned}$$

Likewise, $m_{ip} + m_{iq} = n + |S_i|$ for all $i = 1, \dots, 2^n - 1$. Since these column operations (adding the k th to the j th column and adding the q th to the p th column) lead to two identical columns, we conclude that $\det(M) = 0$ for all $n \geq 3$.

Problem 6 Let $f: [0; +\infty) \rightarrow \mathbb{R}$ be differentiable function such that

$$\int_0^1 f(x) dx = f(1) \text{ and for every } x \geq 1 \text{ holds } xf'(x) + f(x-1) = 0$$

a) Prove that $\lim_{x \rightarrow \infty} f(x) = 0$

b) Determine the radius of convergence of the power series

$$\sum_{n=1}^{\infty} f(n)x^n$$

Asen Bozilov, Stoyan Apostolov (University of Sofia, Bulgaria)

Solution. Rewrite

$$(tf(t))' = f(t) - f(t-1).$$

Fix $x \geq 2$ and integrate from 1 to x :

$$\begin{aligned} xf(x) - f(1) &= \int_1^x f(t)dt - \int_1^x f(t-1)dt = \int_1^x f(t)dt - \int_0^{x-1} f(t)dt = \\ &= \int_{x-1}^x f(t)dt - \int_0^1 f(t)dt. \end{aligned}$$

Hence

$$xf(x) = \int_{x-1}^x f(t)dt,$$

thus, we find $x_1 \in (x-1, x)$, such that

$$f(x) = \frac{f(x_1)}{x}.$$

If $x_1 \geq 2$, apply the same argument to find $x_2 \in (x_1-1, x_1)$, such that

$$f(x) = \frac{f(x_2)}{xx_1}.$$

As long as $x_n \geq 2$, inductively find $x_{n+1} \in (x_n-1, x_n)$, such that

$$f(x) = \frac{f(x_{n+1})}{xx_1x_2 \dots x_n}.$$

Distinguish two cases:

- 1) – the construction could continue indefinitely, i.e. $x_n \geq 2$ for all n . The sequence decreases, let its limit be l . Then

$$|f(x)| = \frac{|f(x_{n+1})|}{xx_1x_2 \dots x_n} \leq \frac{|f(x_{n+1})|}{2^{n+1}} \xrightarrow{n \rightarrow \infty} 0,$$

since $f(x_{n+1}) \rightarrow f(l)$. Thus $f(x) = 0$.

- 2) – the construction could continue only finitely many steps, i.e. for some $n_0 \geq 1$ holds $x_{n_0-1} \geq 2$, but $x_{n_0} < 2$. Since $x_{n+1} \in (x_n - 1, x_n)$, we have $x_{n_0} \geq 1$ and $x_n \geq x - n$ for $n \leq n_0$. Thus $2 > x_{n_0} \geq x - n_0$, hence $n_0 > x - 2$. For $n \leq [x] - 2 < n_0$ we estimate $x_n > [x] - n$, and for $n \in ([x] - 2, n_0]$, $x_n \geq 1$. Then

$$|f(x)| = \frac{|f(x_{n+1})|}{xx_1x_2 \dots x_n} \leq \frac{M}{[x]([x] - 1) \dots 2} \leq \frac{M}{[x]!},$$

where $M = \sup_{x \in [1,2]} |f(x)|$.

As a conclusion, for every $x \geq 2$, $|f(x)| \leq \frac{M}{[x]!}$. The result follow directly.

Individual Competition

Group B

Problem 1 As a monkey was carrying three similar coconuts to the top of a multistory tree, one of the nuts fell to the ground from the 16th floor and crashed. The monkey wants to determine the highest floor for the nut not to crash when dropped to the ground. It can drop a coconut from any of the floors, and pick up if it did not crash. Prove that five trials are enough for the monkey (with two nuts).

Pirmyrat Gurbanov, Gulli Atakishiyeva (International University for the Humanities and Development, Turkmenistan)

Solution:

First trial is to start from the fifth floor. If the coconut did crash, then consequently from the first, second, and then the third floors (if necessary). If after the first trial the coconut is unbroken, then the second trial is to drop it from the ninth floor. If the nut crashed, then drop it from the sixth, seventh, and then the eighth floors (if necessary). If after the second trial the nut is still unbroken, then the third trial is to drop it from the twelfth floor. If crashed, then from the tenth and eleventh floors (if necessary). If unbroken, the fourth trial is to drop the coconut from the fourteenth floor, and next, either from the thirteenth or fifteenth floor.

Problem 2 Given curves $h_1: xy = 2022$ and $h_2: xy = -2022$. Let the straight line $l_1: y = k_1x$ intersect h_1 at the points A and C, and straight line $l_2: y = -k_2x$ intersect h_2 at the points B and D (Where: k_1, k_2 positive reals). Find minimal possible value of the area quadrilateral ABCD.

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution: First, let's find coordinates of A, B, C and D. By intersecting h_1 and l_1 , we will find the coordinates of A and C. Then:

$$\frac{2022}{x} = k_1x \Leftrightarrow x^2 = \frac{2022}{k_1} \Leftrightarrow x = \pm \sqrt{\frac{2022}{k_1}}$$

Now, with the help of x coordinate, let's find the y coordinate of A and C.

$$y = k_1x = k_1 \cdot \left(\pm \sqrt{\frac{2022}{k_1}} \right) = \pm \sqrt{2022k_1}$$

If we repeat the same equations for B and D points, we will have:

$$x = \pm \sqrt{\frac{2022}{k_2}}; y = \pm \sqrt{2022k_2}$$

So, the coordinates of our quadrilateral are:

$$\begin{aligned} A: & \left(-\sqrt{\frac{2022}{k_1}}; -\sqrt{2022k_1} \right) & C: & \left(\sqrt{\frac{2022}{k_1}}; \sqrt{2022k_1} \right) \\ B: & \left(-\sqrt{\frac{2022}{k_2}}; \sqrt{2022k_2} \right) & D: & \left(\sqrt{\frac{2022}{k_2}}; -\sqrt{2022k_2} \right) \end{aligned}$$

Now, let us divide the area of ABCD quadrilateral in to two triangles, BCD and ABD, respectively. We know that area of a triangle with given coordinates of all points is equal to:

$$S = \left| \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \right|$$

Then,

$$S_{BCD} = \left| \frac{1}{2} \begin{vmatrix} \sqrt{\frac{2022}{k_1}} & \sqrt{2022k_1} & 1 \\ \sqrt{\frac{2022}{k_2}} & -\sqrt{2022k_2} & 1 \\ -\sqrt{\frac{2022}{k_2}} & \sqrt{2022k_2} & 1 \end{vmatrix} \right| = \left| \frac{1}{2} \cdot 2022 \begin{vmatrix} \frac{1}{\sqrt{k_1}} & \sqrt{k_1} & 1 \\ \frac{1}{\sqrt{k_2}} & -\sqrt{k_2} & 1 \\ -\frac{1}{\sqrt{k_2}} & \sqrt{k_2} & 1 \end{vmatrix} \right| =$$

$$= \left| \frac{1011}{\sqrt{k_1 k_2}} \cdot \begin{vmatrix} 1 & k_1 \\ 1 & -k_2 \end{vmatrix} \right| = 2022 \left(\sqrt{\frac{k_1}{k_2}} + \sqrt{\frac{k_2}{k_1}} \right)$$

If we repeat the same for ABD triangle we will also get:

$$S_{ABD} = 2022 \left(\sqrt{\frac{k_1}{k_2}} + \sqrt{\frac{k_2}{k_1}} \right)$$

So, our ABCD quadrilateral will have the area:

$$S_{ABCD} = S_{BCD} + S_{ABD} = 2022 \left(\sqrt{\frac{k_1}{k_2}} + \sqrt{\frac{k_2}{k_1}} \right) + 2022 \left(\sqrt{\frac{k_1}{k_2}} + \sqrt{\frac{k_2}{k_1}} \right)$$

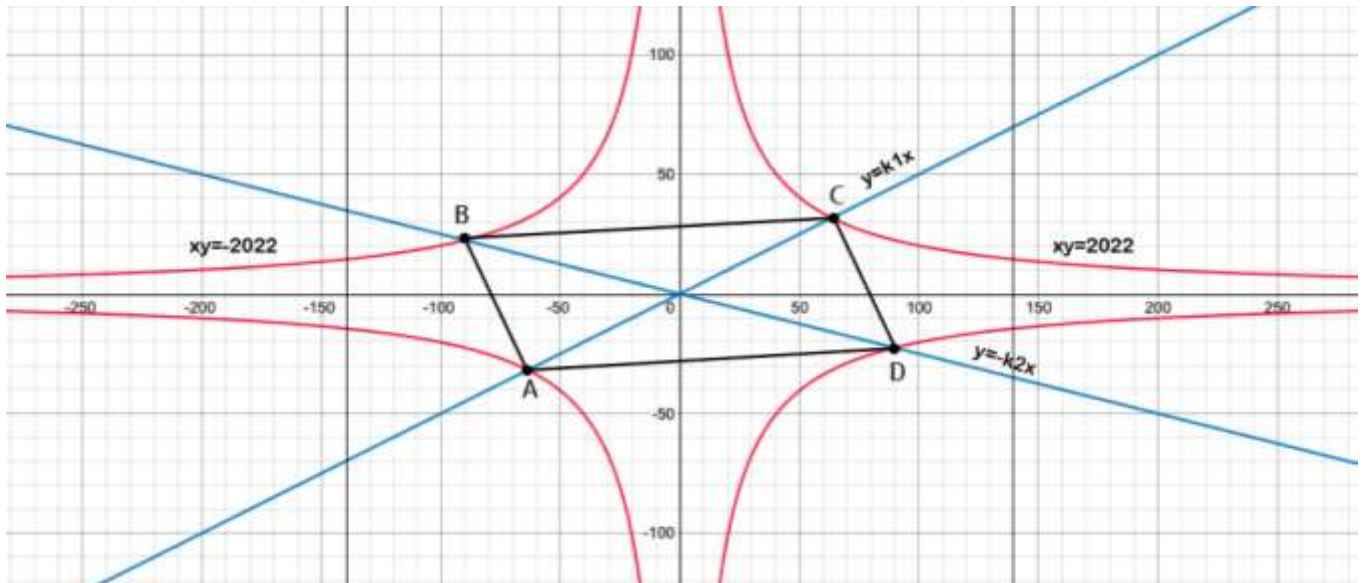
$$S_{ABCD} = 4044 \left(\sqrt{\frac{k_1}{k_2}} + \sqrt{\frac{k_2}{k_1}} \right)$$

Let $t = \frac{k_1}{k_2}$. Then our area will have a function: $f(t) = 4044 \left(\sqrt{t} + \frac{1}{\sqrt{t}} \right)$. Let us derive our function to find minimum value:

$$f'(t) = 4044 \left(\frac{1}{2\sqrt{t}} - \frac{1}{2\sqrt{t^3}} \right) = 0$$

$$\frac{1}{2\sqrt{t}} - \frac{1}{2\sqrt{t^3}} = 0 \Leftrightarrow \frac{1}{2\sqrt{t}} = \frac{1}{2t\sqrt{t}} \Leftrightarrow t = \pm 1 \Rightarrow \frac{k_1}{k_2} = \pm 1$$

We know that k_1 and k_2 are positive reals, then $k_1 = k_2$. Function $f(t)$ has minimal value when l_1 and l_2 are perpendicular. Minimal value of $f(t) = 8088$



Problem 3 Let continuously function $f: [0,1] \rightarrow \mathbb{R}$ such that $\int_0^1 f(x) dx = \frac{\pi}{4}$. Prove that exist $x_0 \in (0,1)$ such that $\frac{1}{1+x_0} < f(x_0) < \frac{1}{2x_0}$.

Yusupbay Ermetov (Turkmen Agricultural Institute, Turkmenistan)

Solution:

Considering $g(t) = \int_0^t \left(f(x) - \frac{1}{1+x^2} \right) dx$ function. It's easy to check $g(0) = g(1) = 0$. By using Mean Value Property exist $x_0 \in (0,1)$ such that $g'(x_0) = 0$. Then

$$f(x_0) = \frac{1}{1+x_0^2}$$

Taking consideration that $x_0 \in (0,1)$ we have

$$x_0 > x_0^2 \Rightarrow 1 + x_0 > 1 + x_0^2 \Rightarrow \frac{1}{1+x_0} < \frac{1}{1+x_0^2}$$

By using AM-GM inequality $1 + x_0^2 > 2x_0$ or $\frac{1}{1+x_0^2} < \frac{1}{2x_0}$

Problem 4 Let $A, B \in M_2(\mathbb{R})$ matrices such that

d) B is invertible

e) $A + B = BA$

f) $\det(A^d - B^d) = \det(A^d + B^d)$ for $d = \{2020; 2021\}$

Prove that $\det(A^{2022} - I_2) = \det(A^{2022} + I_2)$

Murat Chashemov (Turkmenistan)

Solution.

Lemma 1: If $A, B \in M_2(\mathbb{R})$ and $A + B = BA$, then $AB = BA$.

Proof: From condition of lemma

$$(I - B)(I - A) = I - A - B + BA = I = (I - A)(I - B)$$

Then $I - A - B + BA = I - A - B + AB$

From here $AB = BA$.

By using lemma 1 and condition a) we have $AB^{-1} = B^{-1}A$. Multiplying both sides of condition c) to the B^{-d}

$$\det(A^d B^{-d} - I_2) = \det(A^d B^{-d} + I_2)$$

or

$$\det((AB^{-1})^d - I_2) = \det((AB^{-1})^d + I_2)$$

Let $C = AB^{-1}$ and has eigenvalues α and β , then we get

$$(\alpha^d - 1)(\beta^d - 1) = (\alpha^d + 1)(\beta^d + 1)$$

and

$$\alpha^d + \beta^d = 0 \text{ or } \alpha = \beta = 0$$

Taking consideration A and B^{-1} matrices are commute, we know matrices A and B has common eigenvector. Let x is common eigenvector of matrix A and B^{-1} and γ and μ are eigenvalues of A and B^{-1} .

$$AB^{-1}x = \gamma\delta x = 0 \Rightarrow \gamma = 0$$

So A matrix is nilpotent and $A^2 = O_2$

Then $\det(A^{2022} - I_2) = \det(A^{2022} + I_2) = \det(\mp I_2) = 1$

Problem 5 Let $C := 2 \int_0^1 \frac{dt}{\sqrt{t^4+1}}$. Prove that

- a) $\lim_{x \rightarrow 1^-} \sqrt{1-x} \int_0^x \frac{dt}{\sqrt{t^4+(x(1-x))^2}} = C$
b) $\lim_{x \rightarrow 1^-} \frac{C}{\sqrt{1-x}} - \int_0^x \frac{dt}{\sqrt{t^4+(x(1-x))^2}} = 1$

Asen Bozilov, Stoyan Apostolov (University of Sofia, Bulgaria)

Solution.

- a) Making the change of variables $t \rightarrow \sqrt{x(1-x)}$ we obtain

$$\begin{aligned} \sqrt{1-x} \int_0^x \frac{1}{\sqrt{t^4 + (x(1-x))^2}} dt &= \sqrt{1-x} \int_0^{\sqrt{x/(1-x)}} \frac{\sqrt{x(1-x)}}{x(1-x)\sqrt{s^4+1}} ds = \\ &= \frac{1}{\sqrt{x}} \int_0^{\sqrt{x/(1-x)}} \frac{1}{\sqrt{s^4+1}} ds \end{aligned}$$

Letting $x \rightarrow 1$ we obtain that the limit $\int_0^\infty \frac{1}{\sqrt{s^4+1}} ds$.

Making the change of variables $s \rightarrow 1/s$ in $\int_0^1 \frac{1}{\sqrt{s^4+1}} ds$ we obtain the limit C .

- b)

$$\begin{aligned} \frac{C}{\sqrt{1-x}} - \int_0^x \frac{dt}{\sqrt{t^4 + (x(1-x))^2}} &= \\ \frac{1}{\sqrt{1-x}} \int_0^\infty \frac{1}{\sqrt{s^4+1}} ds - \int_0^x \frac{dt}{\sqrt{t^4 + (x(1-x))^2}} &= \\ = \frac{1}{\sqrt{1-x}} \int_0^\infty \frac{1}{\sqrt{s^4+1}} ds - \frac{1}{\sqrt{x(1-x)}} \int_0^{\sqrt{x/(1-x)}} \frac{1}{\sqrt{s^4+1}} ds &= \\ \left(\frac{1}{\sqrt{1-x}} - \frac{1}{\sqrt{x(1-x)}} \right) \int_0^{\sqrt{x/(1-x)}} \frac{1}{\sqrt{s^4+1}} ds + \frac{1}{\sqrt{1-x}} \int_{\sqrt{x/(1-x)}}^\infty \frac{1}{\sqrt{s^4+1}} ds. \end{aligned}$$

As $\lim_{x \rightarrow 1^-} \left(\frac{1}{\sqrt{1-x}} - \frac{1}{\sqrt{x(1-x)}} \right) = 0$, we are left with

$$\lim_{x \rightarrow 1^-} \frac{1}{\sqrt{1-x}} \int_{\sqrt{x/(1-x)}}^\infty \frac{1}{\sqrt{s^4+1}} ds,$$

which could be done by L'Hospital.

Problem 6 Let

$$A(n) = \begin{pmatrix} \frac{1}{1*2} & \frac{1}{2*3} & \cdots & \frac{1}{n(n+1)} \\ \frac{1}{2*3} & \frac{1}{3*4} & \cdots & \frac{1}{(n+1)(n+2)} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{n(n+1)} & \frac{1}{(n+1)(n+2)} & \cdots & \frac{1}{(2n-1)*2n} \end{pmatrix}, \quad B(n) = \begin{pmatrix} 0! & 1! & \cdots & (n-1)! \\ 1! & 2! & \cdots & n! \\ \vdots & \vdots & \ddots & \vdots \\ (n-1)! & n! & \cdots & (2n-2)! \end{pmatrix}$$

for all $n \geq 1$. Evaluate

$$\frac{\det(B(n))}{\det(A(n))}$$

Robert Skiba (Nicolaus Copernicus University in Torun, Poland)

Solution:

It is clear that for $n = 1$,

$$\frac{\det(B(1))}{\det(A(1))} = 2.$$

Fix $n \geq 2$. For any $0 \leq t \leq n-1$ we define inductively A_t as follows

$$A_0 := A(n) \text{ for } t = 0 \text{ and } A_t = \frac{1}{t+1} X_{t-1} A_{t-1} \text{ for } t \geq 1,$$

Where

$$X_t = \begin{bmatrix} 1 & -1 & 0 & \cdots & & & 0 \\ 0 & 1 & -1 & 0 & \ddots & & \\ \vdots & 0 & \ddots & \ddots & & & \\ & & \ddots & -1 & 0 & & \\ & & & 1 & -1 & 0 & \\ & & & 0 & t+2 & 0 & \\ & & & & 0 & t+2 & \vdots \\ & & & & & \ddots & \ddots & 0 \\ 0 & & & & & \cdots & 0 & t+2 \end{bmatrix}$$

Is a squared matrix of dimension n and the number $t+2$ appears $t+1$ times in the matrix X_t , for $0 \leq t \leq n-1$. Since

$$(t+2)A_{t+1} = X_t A_t \text{ for } 0 \leq t \leq n-1,$$

it follows that

$$(t+2)A_{t+1}[i,j] = \begin{cases} A_t[i,j] - A_t[i+1,j] & \text{if } i < n-t, \\ (t+2)A_t[i,j] & \text{if } i \geq n-t, \end{cases} \quad (2)$$

where $A_t[i,j]$ stands for the entry in the i -th row and j -th column of a matrix A_t . We shall prove by mathematical induction that

$$A_t[i,j] = \begin{cases} \frac{(i+j-2)!}{(i+j+t)!} & \text{if } i < n-t, \\ \frac{(i+j-2)!}{(j+n)!} & \text{if } i \geq n-t, \end{cases} \quad (3)$$

for all $0 \leq t \leq n-1$.

Base case: We show that the statement (3) holds for $t=0$. Indeed, we have

$$A_0[i,j] = A(n)[i,j] = \frac{1}{(i+j-1) \cdot (i+j)} = \frac{(i+j-2)!}{(i+j)!}.$$

Inductive step: We assume that (3) holds for t such that $0 \leq t \leq n-2$.

We will show that (3) also holds for $t+1$. This will be proceeded in three cases:

Case 1: If $i \geq n-t > n-(t+1)$, then, in view of (2) and (3), we have

$$(t+2)A_{t+1}[i,j] = (t+2)A_t[i,j] = (t+2) \frac{(i+j-2)!}{(j+n)!}.$$

Case 2: If $i = n-(t+1)$, then, in view of (2) and (3), we have

$$\begin{aligned} (t+2)A_{t+1}[i,j] &= A_t[i,j] - A_t[i+1,j] = \frac{(i+j-2)!}{(i+j+t)!} - \frac{(i+j+1-2)!}{(j+n)!} = \\ &= (j+n) \frac{(i+j-2)!}{(j+n)!} - (i+j-1) \frac{(i+j-2)!}{(j+n)!} = (n-i+1) \frac{(i+j-2)!}{(j+n)!} = \\ &= (t+2) \frac{(i+j-2)!}{(j+n)!}. \end{aligned}$$

Case 3: If $i < n-(t+1)$, then in view of (2) and (3), we have

$$\begin{aligned} (t+2)A_{t+1}(i,j) &= A_t[i,j] - A_t[i+1,j] = \frac{(i+j-2)!}{(i+j+t)!} - \frac{(i+j-1)!}{(i+j+t+1)!} = \\ &= (i+j+t+1) \frac{(i+j-2)!}{(i+j+t+1)!} - (i+j-1) \frac{(i+j-2)!}{(i+j+t+1)!} = \\ &= (t+2) \frac{(i+j-2)!}{(i+j+(t+1))!}. \end{aligned}$$

This proves (3). Consequently, for $t = n-1$ we get

$$A_{n-1}[i, j] = \frac{(i + j - 2)!}{(j + n)!}.$$

Let D be a diagonal matrix given by

$$D = \text{diag}((n + 1)!, (n + 2)!, \dots (n + n)!).$$

Then

$$\begin{aligned} B(n) &= A_{n-1}D = \frac{1}{n}X_{n-2}A_{n-2}D = \frac{1}{n \cdot (n-1)}X_{n-2}X_{n-3}A_{n-3}D = \dots \\ &= \frac{1}{n!}X_{n-2}X_{n-3} \dots X_0A_0D. \end{aligned}$$

Hence

$$\begin{aligned} \det(B(n)) &= \frac{1}{(n!)^n} \det(X_{n-2}) \det(X_{n-3}) \dots \det(X_0) \det(A_0) \det(D) = \\ &= \frac{1}{(n!)^n} n^{n-1} (n-1)^{n-2} \dots 2^1 (n+1)! (n+2)! \dots (n+n)! \det(A_0) = \\ &= \frac{(n+1)!}{1!} \frac{(n+2)!}{2!} \dots \frac{(n+n)!}{n!} \det(A_0) = \\ &= (n!)^n \binom{n+1}{1} \binom{n+2}{2} \dots \binom{2n}{n} \det(A(n)). \end{aligned}$$

Finally, we get

$$\frac{\det(B(n))}{\det(A(n))} = (n!)^n \binom{n+1}{1} \binom{n+2}{2} \dots \binom{2n}{n}.$$

This complete the solution.

Team Competition

Problem 1 Let a real number and $(x_n)_{n \geq 0}$ numerical sequence satisfying conditions $x_0 = 0$, $x_1 = \frac{1}{2022}$ and $x_{n+1} = ax_n + 2021x_{n-1}$ for every $n \geq 1$. Prove that:

$$\sum_{n=1}^{\infty} \frac{1}{x_n^2 - x_{n-1}x_{n+1}}$$

numerical series convergent and find it's limit.

Yusupbay Ermetov (Turkmen Agricultural Institute, Turkmenistan)

Solution Let $y_n = x_n^2 - x_{n-1}x_{n+1}$ for every $n \geq 1$.

If $n = 1$ we have $y_1 = x_1^2 - x_0x_2 = \frac{1}{2022^2}$

Using given conditions of problem it's easy to rewrite as matrix equality

$$\begin{pmatrix} a & 2021 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x_{n-1} & x_n \\ x_{n-2} & x_{n-1} \end{pmatrix} = \begin{pmatrix} x_n & x_{n+1} \\ x_{n-1} & x_n \end{pmatrix}$$

Taking determinant from both sides of equality

$$y_n = x_n^2 - x_{n-1}x_{n+1} = -2021(x_{n-1}^2 - x_{n-2}x_n) = -2021y_{n-1} = (-1)^{n-1}2021^{n-1}y_1$$

Then $y_n = \frac{1}{2022^2} (-1)^{n-1} 2021^{n-1}$

$$\sum_{n=1}^{\infty} \frac{1}{x_n^2 - x_{n-1}x_{n+1}} = \sum_{n=1}^{\infty} \frac{2022^2}{(-2021)^{n-1}} = 2022^2 \frac{1}{1 + \frac{1}{2021}} = 4086462$$

Problem 2 Find all a, b, c positive integers such that $a \mid b^2 \mid c^3 \mid a^4 \mid b^5 \mid c^6 \dots$

Where $a \mid b$ means a divides b .

Kerven Durdymyradov (The State Energy Institute of Turkmenistan, Turkmenistan)

Solution:

Let p^x, p^y and p^z are exact powers of a prime p that *divide* a, b and c respectively.

By condition of problem $a^{3k+1} \mid b^{3k+2} \mid c^{3k+3} \mid a^{3k+4} \mid b^{3k+5} \mid c^{3k+6}$ for every k non-negative integer. From $a^{3k+1} \mid b^{3k+2}$ and $b^{3k+2} \mid a^{3k+4}$ we get

$$\begin{cases} (3k+2)y \geq (3k+1)x \\ (3k+4)x \geq (3k+2)y \end{cases} \Rightarrow \begin{cases} y \geq \frac{3k+1}{3k+2}x \\ x \geq \frac{3k+2}{3k+4}y \end{cases} \text{ for sufficiently large } k \text{ we get } x = y \text{ for}$$

every p prime.

Similarly way $x = y = z$ for every p prime. Then we get $a = b = c$.

Answer $(a, b, c) \in (k, k, k)$ where $k \in \mathbb{N}$

Problem 3 Two differentiable real functions f and g satisfy

$$\frac{f'(x)}{g'(x)} = e^{f(x)-g(x)}$$

for all x and $f(0) = g(2022) = 1$. Find the largest constant c such that $f(2022) > c$ for all such functions f, g .

Gurtmyrat Motukov (Magtymguly Turkmen State University, Turkmenistan)

Solution. The main hypothesis may be rewritten, equivalently, $(e^{-f(x)})' = (e^{-g(x)})'$ for all $x \in \mathbb{R}$. Thus $h(x) := e^{-f(x)} - e^{-g(x)} = \text{const}$. Writing $h(0) = h(2021)$ we obtain, by hypothesis,

$$e^{-1} - e^{-g(0)} = e^{-f(2021)} - e^{-1}.$$

Thus, $e^{-f(2021)} = 2e^{-1} - e^{-g(0)} < 2e^{-1} = e^{-1+\ln 2}$. This implies that the largest constant c such that $f(2021) > c$ for all such functions f, g is $c = 1 - \ln 2$.

Problem 4 Calculate limit

$$\lim_{n \rightarrow \infty} n \int_0^1 x^n \arctg x \, dx$$

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution: Let f be a continuous, real valued function on $[0; 1]$. Then

$$\lim_{n \rightarrow \infty} n \int_0^1 x^n f(x) \, dx = f(1)$$

Proof: Let's to prove lemma when $f(x) = x^k$

$$\lim_{n \rightarrow \infty} n \int_0^1 x^n x^k \, dx = \lim_{n \rightarrow \infty} n \frac{1}{n+k+1} = 1 = f(1)$$

So it's easy to check for $f(x) = p_k(x) = a_0 + a_1x + a_2x^2 + \dots + a_kx^k$

$$\begin{aligned} \lim_{n \rightarrow \infty} n \int_0^1 x^n p_k(x) \, dx &= \lim_{n \rightarrow \infty} n \int_0^1 x^n \sum_{i=0}^k a_i x^i \, dx = \lim_{n \rightarrow \infty} \sum_{i=0}^k a_i n \left(\int_0^1 x^i x^n \, dx \right) \\ &= \sum_{i=0}^k a_i \lim_{n \rightarrow \infty} \left(n \left(\int_0^1 x^i x^n \, dx \right) \right) = \sum_{i=0}^k a_i = p_k(1) = f(1) \end{aligned}$$

For general proof we will use Stone-Weirstrasse theorem

Theorem: Suppose f is a continuous real-valued function defined on the real interval $[a, b]$. For every $\varepsilon > 0$, there exists a polynomial p such that for all x in $[a, b]$, we have $|f(x) - p(x)| < \varepsilon$

$$\begin{aligned} \left| n \int_0^1 x^n f(x) \, dx - f(1) \right| &= \left| n \int_0^1 x^n f(x) \, dx - n \int_0^1 x^n p_k(x) \, dx + n \int_0^1 x^n p_k(x) \, dx - p_k(1) + p_k(1) - f(1) \right| \\ &\leq \left| n \int_0^1 x^n f(x) \, dx - n \int_0^1 x^n p_k(x) \, dx \right| + \left| n \int_0^1 x^n p_k(x) \, dx - p_k(1) \right| + |p_k(1) - f(1)| \\ &< \frac{2\varepsilon}{3} + \left| n \int_0^1 x^n (f(x) - p_k(x)) \, dx \right| < \frac{2\varepsilon}{3} + n \int_0^1 x^n |f(x) - p_k(x)| \, dx < \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} n \int_0^1 x^n \, dx \\ &= \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} n \frac{1}{n+1} < \varepsilon \end{aligned}$$

We have

$$\lim_{n \rightarrow \infty} n \int_0^1 x^n f(x) \, dx = f(1)$$

Function $f(x) = \arctg x$ satisfies the assumptions of the lemma. Then

$$\lim_{n \rightarrow \infty} n \int_0^1 x^n \arctg x \, dx = \arctg 1 = \frac{\pi}{4}$$

Problem 5 Let us consider a cube of side length 2. Let S_1 be a circle circumscribed about the bottom face, and let S_2 be a circle inscribed in the side face of the cube. Find the distance between the circles S_1 and S_2 , i.e. the minimal distance between points of those circles.

Marcin Zygmunt (University of Silesia, Poland)

Solution: Let us consider two spheres with center at the center of the cube with each containing one of the circles mentioned in the problem. The distance between the circles cannot clearly be less than the distance between the spheres, which is equal of course to the difference between their radii. On the other hand, it is easy to see that the cone formed of rays from the center and passing through one circle, intersects the second circle, hence there is a ray from the center that passes through both circles. It follows that this distance is minimal.

Let us now put the whole picture in the system of Cartesian coordinates in a such way that the center of the cube is at origin $(0, 0, 0)$ and its vertices are at points $(\pm 1, \pm 1, \pm 1)$. So, the spheres are also centered at the origin. The inner circle touches the center of side of the cube, i.e. it passes through the point $(1, 1, 0)$, hence it has the radius $\sqrt{2}$. The outer sphere contains instead vertices of the cube, so it has the radius $\sqrt{3}$. Thus the difference between radius of the spheres, hence the minimal distance between points of circles S_1 and S_2 , is equal $\sqrt{3} - \sqrt{2}$.

Problem 6 Let n be a positive integer and let $A, B \in M_n(C)$ such that $A^2 = A$ and $B^2 = B$ then show that

$$\text{rank}(A - AB) + \text{rank}(AB - B) = \text{rank}(A - B)$$

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution:

Easy to see $A(A - B)B = A^2B - AB^2 = AB - AB = 0$

By using Frobenius inequality:

$$\text{rank}(UV) + \text{rank}(VW) \leq \text{rank}(UVW) + \text{rank}(V)$$

If $U = A$, $V = A - B$ and $W = B$ then we get

$$\begin{aligned} \text{rank}(A(A - B)) + \text{rank}((A - B)B) &= \text{rank}(A^2 - AB) + \text{rank}(AB - B^2) \\ &= \text{rank}(A - AB) + \text{rank}(AB - B) \leq \text{rank}(A(A - B)B) + \text{rank}(A - B) \\ &= \text{rank}(A - B) \end{aligned}$$

On the other hand using $\text{rank}(U) + \text{rank}(V) \geq \text{rank}(U + V)$ for $U = A - AB$, and $V = AB - B$

$$\text{rank}(A - AB) + \text{rank}(AB - B) \geq \text{rank}(A - B)$$

So taking consideration both inequalities we get necessary equality.

Problem 7 Let $\xi_1, \xi_2, \xi_3, \dots, \xi_n, \dots$ be sequence of identically distributed continuously independent random variables and $\nu = \min\{k: \xi_k > \xi_1\}$. Find mathematical expectation $E(\nu)$.

Orazmamet Annaorazov (Magtymguly Turkmen State University, Turkmenistan)

Solution:

Since the independent random variables $\xi_1, \xi_2, \xi_3, \dots, \xi_n, \dots$ are continuous, therefore $P\{\xi_i = \xi_j\} = 0$ for $i \neq j$. Let's write the law of distribution of a discrete random variable ν

$$\begin{array}{ccccccc} \nu & 2 & 3 & & & n & \\ P & P_2 & P_3 & \cdots & \cdots & P_n & \cdots \end{array}$$

Where $P_n = P\{\nu = n\}$.

Taking consideration that $P\{\xi_1 = \max(\xi_1; \xi_2; \dots; \xi_n)\} = \frac{1}{n}$

Then,

$$\begin{aligned} P_n &= P\{\nu = n\} = P\{\xi_1 = \max(\xi_1; \xi_2; \dots; \xi_{n-1})\} - P\{\xi_1 = \max(\xi_1; \xi_2; \dots; \xi_n)\} = \\ &= \frac{1}{(n-1)} - \frac{1}{n} = \frac{1}{n(n-1)} \end{aligned}$$

So,

$$M\nu = \sum_{n=2}^{\infty} nP_n = \sum_{n=2}^{\infty} n \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \frac{1}{(n-1)} = \infty$$

Problem 8 Calculate the exact value of the integral

$$\int_0^{+\infty} \frac{(1 - \cos x)^2}{x^3} dx$$

Pirmyrat Gurbanov (International University for the Humanities and Development, Turkmenistan)

Solution: Answer: $\ln 2$

We will use the following theorem

Theorem (Frullani's integral). Let $f : R_+ \rightarrow R$ be a continuous, real valued function such that there exist limits $f_0 = \lim_{x \rightarrow 0+} f(x)$ and $f_\infty = \lim_{x \rightarrow +\infty} f(x)$. Then for every $\alpha, \beta \in R_+$ the following holds

$$\int_0^{+\infty} \frac{f(\alpha x) - f(\beta x)}{x} dx = (f_\infty - f_0) \ln \left(\frac{\alpha}{\beta} \right)$$

We have

$$\begin{aligned} \frac{\frac{1 - \cos x}{x^2} - \frac{1 - \cos 2x}{4x^2}}{x} &= \frac{4 - 4 \cos x - 1 + \cos 2x}{4x^3} = \frac{4 - 4 \cos x - 1 + 2 \cos^2 x - 1}{4x^3} \\ &= \frac{2 - 4 \cos x + 2 \cos^2 x}{4x^3} = \frac{(1 - \cos x)^2}{2x^3} \end{aligned}$$

Function $f(x) = \frac{1 - \cos x}{x^2}$ satisfies the assumptions of the theorem with $f_0 = \frac{1}{2}, f_\infty = 0$

The integral in question is Frullani's integral with $\alpha = 1, \beta = 2$.

$$\begin{aligned} \int_0^{+\infty} \frac{(1 - \cos x)^2}{x^3} dx &= 2 \int_0^{+\infty} \frac{(1 - \cos x)^2}{2x^3} dx = 2 \int_0^{+\infty} \frac{\frac{1 - \cos x}{x^2} - \frac{1 - \cos 2x}{4x^2}}{x} dx \\ &= 2 \left(0 - \frac{1}{2} \right) \ln \left(\frac{1}{2} \right) = \ln 2 \end{aligned}$$

Problem 9 Find all differentiable functions $f: R \rightarrow R$ such that for all $x, y \in R$ the inequality $f(x + y) \geq 2022^x f(y) + f(x)$ holds and $f(0) = 0, f(1) = 1$.

Yagub Aliyev (ADA University, Azerbaijan)

Solution: Rewrite the inequality as $f(x + y) - f(x) \geq 2022^x f(y)$. Let $y > 0$. By dividing both sides of this inequality to y we obtain $\frac{f(x + y) - f(x)}{y} \geq 2022^x \frac{f(y) - f(0)}{y - 0}$. By tending $y \rightarrow 0$ we obtain $f'(x) \geq 2022^x f'(0)$. Now, suppose that $y < 0$. In this case, by dividing both sides of this inequality to y we obtain $\frac{f(x + y) - f(x)}{y} \leq 2022^x \frac{f(y) - f(0)}{y - 0}$. By tending $y \rightarrow 0$ we obtain $f'(x) \leq 2022^x f'(0)$. It follows that $f'(x) = 2022^x f'(0)$. If $x > 0$ then the equality can be written as $\frac{f(x + y) - f(y)}{x} \geq f(y) \frac{2022^x - 1}{x} + \frac{f(x) - f(0)}{x - 0}$. By tending $x \rightarrow 0$ we obtain $f'(y) \geq f(y) \ln 2022 + f'(0)$. Similarly, if $x < 0$ then we obtain $f'(y) \leq f(y) \ln 2022 + f'(0)$. Therefore $f'(y) = f(y) \ln 2022 + f'(0)$. Using the above equality $f'(y) = 2022^y f'(0)$ we obtain that $f(y) \ln 2022 + f'(0) = 2022^y f'(0)$. Setting $y = 1$ we obtain $f'(0) = (\ln 2022) / 2021$. Then $f(x) = (2022^x - 1) / 2021$ for all x .

Problem 10 Let V_n be the id number of the competitor who won or shall win the n^{th} OMOUS Competition. Show that there is a positive real number Ω (called Open Mathematical Olympiad's Constant) such that the number $\left[10^{\cdot^{10^\Omega}}\right]_{n-times}$ in decimal representation starts with number V_n followed with zeroes for each positive integer n . (Note: we assume that the number of participant is limited, i.e. for instance it'll never overpass one million).

Marcin Zygmunt (University of Silesia, Poland)

Solution. There is a such real number for any bounded sequence of positive integers.

Let us assume that the biggest number in sequence V_n has M digits. We form now another sequence v_n in the following way. We start with $v_1 = V_1 10^M$.

Then $v_n = V_n \cdot 10^{\kappa_n}$, where $\kappa_n = [v_{n-1} - \log_{10} V_n]$ is taken in a such way, that

$$\frac{1}{V_n} 10^{v_{n-1}} \leq 10^{\kappa_n} < \frac{1}{V_n} 10^{v_{n-1}+1}.$$

Hence

$$10^{v_{n-1}} \leq v_n = V_n \cdot 10^{\kappa_n} < v_n + 1 \leq 10^{v_{n-1}+1}$$

or equivalently

$$v_{n-1} \leq \log_{10} v_n < \log_{10}(v_n + 1) \leq v_{n-1} + 1. \quad (1)$$

Let now

$$p_n = \log_{10}(\log_{10}(\dots(\log_{10} v_n) \dots))$$

And

$$q_n = \log_{10}(\log_{10} \dots (\log_{10}(v_n + 1)) \dots)$$

Hence

$$p_{n-1} \leq p_n < q_n \leq q_{n-1}$$

From inequality (1). So, the sequence (p_n) is monotone and bounded, so it is convergent. Finally, we put

$$\Omega = \lim_{n \rightarrow \infty} p_n.$$

Remark. The same construction works for much wider class of integer sequences, they do not need to be bounded. For instance, one can show that there is a positive real number c such that $\left[2^{\cdot^{2^c}}\right]_{n-times}$ is a prime number for all positive integers n .